

BEYOND THE HORIZON

SKYGAZE INDIA

Also Inside: Astronomical Events | Astrophotographs Lunar
Calendar | The Stellar Life Cycle | Lunar Eclipse | Exoplanets

SEPTEMBER 2025 | VOL. 1, ISSUE 3

INTRODUCTION

Beyond the Horizon, Skygaze India's monthly publication, is meticulously curated to stimulate intellectual curiosity and foster deeper public engagement with astronomy and space science.

Each issue presents an amalgamation of celestial event highlights, exemplary astrophotography features, perspicacious articles, and exclusive content contributed by our community of astronomy enthusiasts. Our objective is to make space science both accessible and profoundly inspiring.

OUR TEAM

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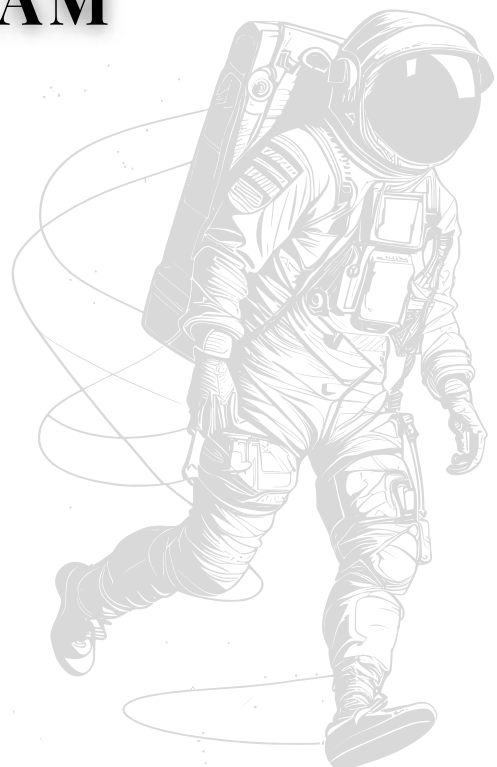
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"At Skygaze India, our mission is to make space science accessible—to bring the cosmos closer by turning complex knowledge into everyday curiosity for all."

Harsh
Founder & CEO

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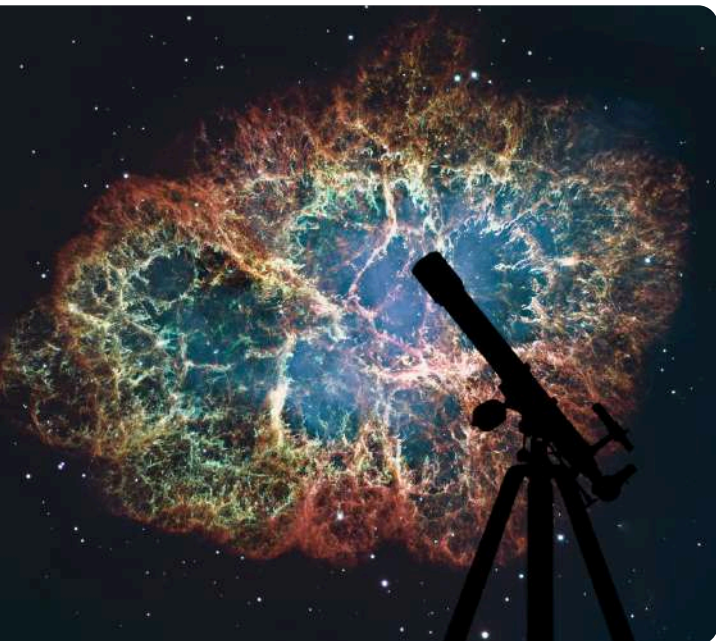
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ASTRONOMICAL EVENTS



SEPTEMBER 7: HARVEST FULL MOON

The Harvest Moon is the full moon closest to the autumn equinox. It will shine brightly all night, giving farmers extra light to work under in ancient times. This phenomenon is cherished by stargazers for its golden hue.

SEPTEMBER 7: TOTAL LUNAR ECLIPSE

During this eclipse, Earth will pass between the Sun and Moon, casting a shadow over the Moon and giving it a reddish glow — visible from many parts of the world.

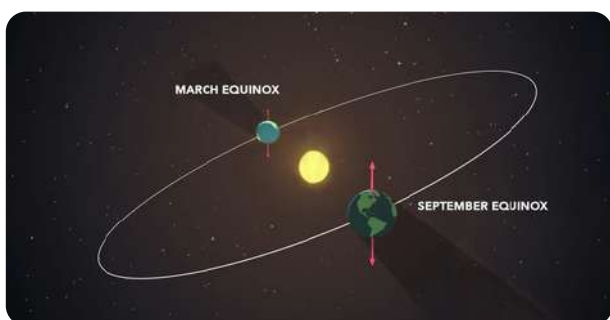


SEPTEMBER 12: CLOSE CONJUNCTION OF PLEIADES AND MOON

The Moon will pass close to the Pleiades star cluster, creating a beautiful sight in the night sky. Best viewed with binoculars for extra detail and perfect for astrophotography.

SEPTEMBER 21: SATURN OPPOSITION

Saturn will be directly opposite the Sun from Earth's viewpoint, making it appear brighter and clearer than usual. A telescope will easily reveal its stunning planetary rings.



SEPTEMBER 22: AUTUMN EQUINOX

Day and night will be almost equal in length worldwide as the Sun crosses the celestial equator. Marks the first day of autumn in the Northern Hemisphere. It's a symbolic time of balance and change.



SEPTEMBER 22: NEW MOON

The Moon will not be visible in the sky, creating the darkest nights of the month — perfect for stargazing and spotting faint celestial objects. This is also an ideal time for deep-sky astrophotography.

SEPTEMBER 23: NEPTUNE OPPOSITION

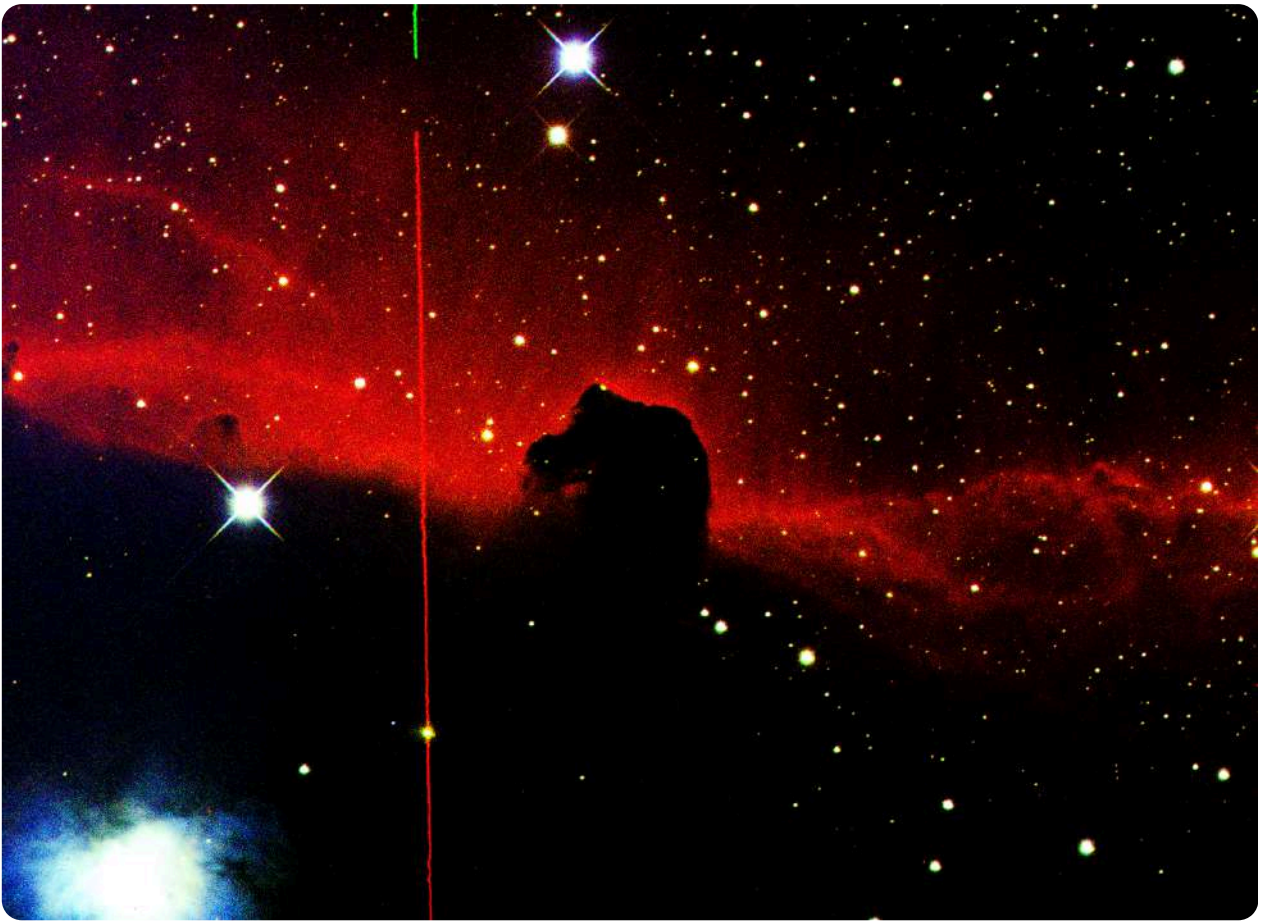
Neptune will be at its closest and brightest for the year, though it's still too faint to see without a telescope. Under good viewing conditions, its subtle blue color becomes visible.



LUNAR CALENDAR

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	01  64.39% First Quarter 8.75 day	02  73.51% Waxing Gibbous 9.68 day	03  81.93% Waxing Gibbous 10.64 day	04  89.24% Waxing Gibbous 11.62 day	05  94.99% Waxing Gibbous 12.64 day	06  98.71% Full Moon 13.7 day
07  100% Full Moon 14.78 day	08  98.59% Full Moon 15.88 day	09  94.45% Waning Gibbous 17 day	10  87.78% Waning Gibbous 18.12 day	11  79.03% Waning Gibbous 19.24 day	12  68.77% Last Quarter 20.34 day	13  57.67% Last Quarter 21.42 day
14  46.35% Last Quarter 22.49 day	15  35.4% Last Quarter 23.54 day	16  25.33% Waning Crescent 24.57 day	17  16.59% Waning Crescent 25.59 day	18  9.49% Waning Crescent 26.59 day	19  4.3% Waning Crescent 27.57 day	20  1.12% New Moon 28.53 day
21  0% New Moon 29.48 day	22  0.88% New Moon 0.88 day	23  3.62% New Moon 1.8 day	24  8.06% Waxing Crescent 2.71 day	25  13.98% Waxing Crescent 3.6 day	26  21.16% Waxing Crescent 4.49 day	27  29.36% Waxing Crescent 5.38 day
28  38.35% First Quarter 6.28 day	29  47.87% First Quarter 7.18 day	30  57.67% First Quarter 8.11 day				

ASTROPHOTOGRAPHS



Horsehead Nebula



Omega Centauri



Messier 6

Captured and processed by:
Mr. Gukuu Bingunmeh Justpa
Student, University of Energy and Natural Resources, Ghana

THE STELLAR LIFE CYCLE: BIRTH TO COSMOS RECYCLING

Introduction:

Stellar evolution is considered as one of the most crucial topics in astrophysics, as stars are the brilliant engines of our universe (Salaris and Cassisi, 2005). The stellar **mass** dictates every stage of its evolution, where they undergo an extraordinary transformation from birth to death. The dynamic interplay between **gravity** and **nuclear fusion** is the reflection of their life cycle which spans million to trillions of years. This cycle is culminating in tragedian endpoint, whether a supernova formation or fading into a stellar remnant (Prialnik, 2000). The profound connection between stellar processes and the cosmos recycling of matter is clearly understood through this wonderful journey.

Star Formation: Nebulae to Protostars

The very starting point of this journey is the presence of a **molecular cloud** called **nebula**, where stars begin their lives (Burkhart et al., 2024). These nebulae, contain the raw materials for the star formation, such as the Orion and Taurus-constellations (Koda and Tan, 2025). The Gravitational forces cause the dust and gases in those regions to fragment and collapse leading to the formation of what we call a **protostar**; The protostar undergoes further contraction and develops strong stellar wind that expels some of its material away from the surface producing what is called a **T-Tauri phase**. However, a **true star** is not born yet, it requires a temperature of 15 MK to ignite a nuclear fusion

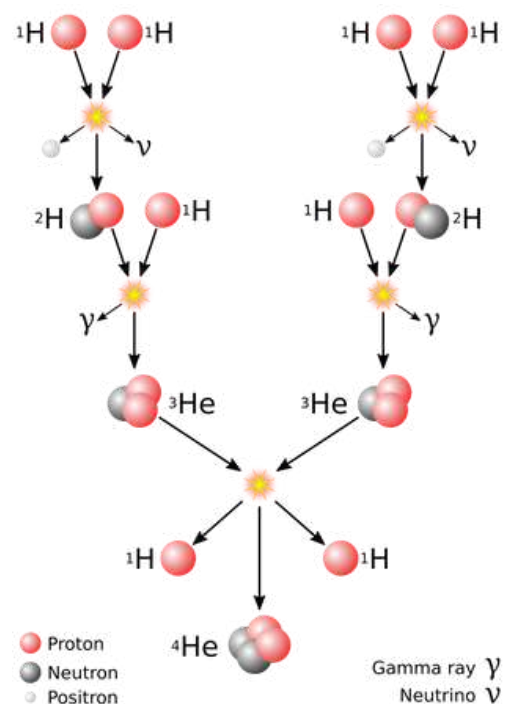
(Keilmann, 2025). To reach this point, the protostar accumulates matter from surrounding regions and contract leading to heating up, this marks the birth of a true star at the mentioned temperature above, and hydrogen burning starts to happen forming helium and releases energy (heat and light).

Main Sequence: Stability

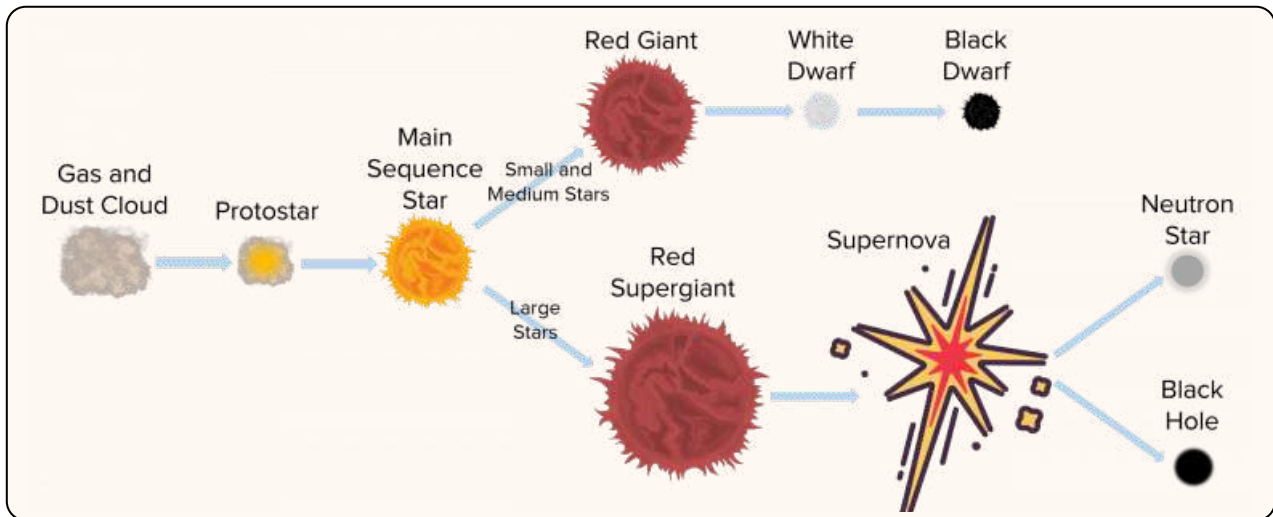
The next stage is characterized by a stable balance between the inward gravity and the radiation pressure (outward force from fusion) which is called the **proton -proton chain** with an overall reaction as follows:



the longest stage in the star's life is constituted in this phase with a range from millions of years for a massive star to trillions of years for red dwarf of low mass star.



The Proton-Proton chain, Wikipedia



Stellar life cycle diagram, Tutoroot

Post-Main Sequence Evolution

• Red Giant phase

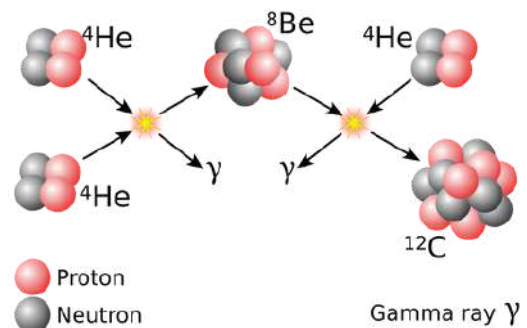
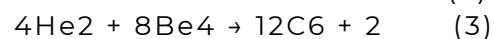
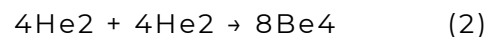
The hydrogen is exhausted now, and the fusion reaction ceases, this causes the core to contract in addition to the expansion of the outer layers. Average stars in mass forms a **red giant**, while massive stars turn into a red **supergiant**. This phase is characterized by burning the hydrogen and the accumulation of helium in the core.

• Helium Flash

This process starts under certain conditions in stars that ranges between 0.8 and 2.0 solar masses, where the helium core becomes in a state as quantum pressure supports it against collapse (Hansen et al., 2004). This state is named "**electron-degenerate**". At a temperature of 100 million Kelvin, the helium fusion ignites with explosives in a **Helium Flash**.

The helium flash is considered as a runaway reaction where helium fuses into carbon via a process called "**triple-alpha process**" presented in equations 2 & 3. This process releases energy equivalent to Milky Way's output for very short time not exceeding few seconds.

After the helium flash the star transitions to stable helium burning, as the core expands and cools.



The Helium Flash triple alpha diagram, Wikipedia

For stars with masses > 2.25 , helium encounters a smooth ignition not expressing any flash as it occurs without degeneracy.

For stars with masses < 0.5 solar masses, they evolve into white dwarfs as helium fusion is never achieved.

Star Death: The Role of Mass

• Low & intermediate stars

These stars expand into Red Giants stars, their shells fuses hydrogen and helium. The encounter thermal pulses and strong wind that eject the outer layer forming "planetary nebula".



Hubble's Crab Nebula, Esahubble

The remaining core turns into **white dwarf** (a dense Earth sized object 10^6 ton/cm^3). This white dwarf cools over time leading to the formation of a **black dwarf** (Ryan and Norton, 2010).

Massive stars

These stars evolve rapidly, where the fusion accumulates producing several sequenced elements (Hydrogen – helium – carbon – oxygen – neon – magnesium). Not only that, for largest stars, the chain continues until silicon fuses into Iron. The iron fusion consumes huge energy; therefore, it stops the fusion and starts to collapse. The core rebounds and leads to a **supernova explosion** that disperse heavy elements into the space.

The remnant materials become either a **neutron star** or, a **black hole** for cores with three times of solar masses.

Key Mass Thresholds:

- Below $0.5 M_{\odot}$: Stars form helium white dwarfs.
- $0.5\text{--}8 M_{\odot}$: Stars produce white dwarfs via planetary nebulae.
- $8\text{--}20 M_{\odot}$: Stars leave behind neutron stars.

Above $20 M_{\odot}$: Stars collapse into black holes.

Cosmic recycle and conclusion

The interstellar medium is rich with heavy elements resulted from stellar death. This enables the formation of new stars, planets. The carbon, oxygen and iron are provided to galaxies through Supernovae and planetary nebulae, those elements are crucial for planetary and biological systems.

In conclusion, Gazing at the sky, illustrates that we witness various stellar stages, telling a story of renewal and underscores a connection to the cosmos.

-Mina Gerges

Department of Physics, College of Science, United Arab Emirates University,
UAE

LUNAR ECLIPSE: WHEN THE EARTH STEALS THE SPOTLIGHT

On most nights, the Moon bathes in the Sun’s reflected glow, lighting up our skies in a soft, silvery hue. But once in a while, the Earth steps in, casting a shadow that turns this celestial spotlight into a cosmic drama — the **Lunar Eclipse**.

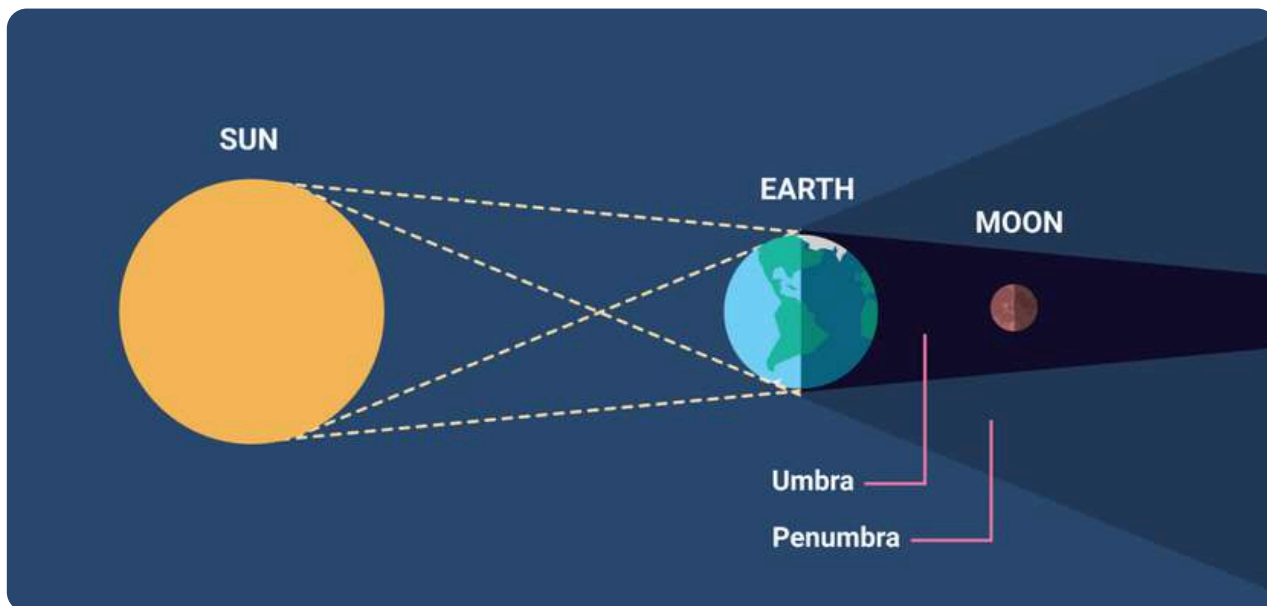
A lunar eclipse occurs when the **Earth comes directly between the Sun and the Moon**, blocking sunlight from reaching the Moon. This alignment can only happen during a **Full Moon**, and it results in the Earth casting its shadow — known as the **umbra** — on the lunar surface.

But unlike solar eclipses, lunar eclipses are **safe to view with the naked eye** and can be seen from **anywhere on the night side of Earth**, making them more accessible and awe-inspiring for skywatchers across the globe.

Not all lunar eclipses are created equal. Based on the degree of Earth’s shadow covering the Moon, there are three main types:

- **Penumbral Lunar Eclipse:** The Moon passes through the Earth’s outer shadow (called penumbra). The Moon looks just a little bit dim. Many people don’t even notice it.
- **Partial Lunar Eclipse:** A part of the Moon goes into the Earth’s darker shadow (called umbra), making part of the Moon look clearly dark.
- **Total Lunar Eclipse:** The entire Moon goes into the umbra. During this, the Moon turns a reddish or copper color. That’s why it’s also called a “Blood Moon”.





Total Lunar Eclipse

Why Does the Moon Turn Red?

During a total eclipse, sunlight still reaches the Moon — but only after being filtered through Earth's atmosphere. Shorter wavelengths (like blue) scatter away, while longer **red and orange wavelengths bend around Earth and illuminate the Moon**. This same phenomenon makes our sunsets and sunrises appear reddish.

The exact shade of red depends on atmospheric conditions. For instance, **volcanic eruptions or high pollution** can intensify the color, creating a hauntingly beautiful Moon.

Cultural Significance Through the Ages

For centuries, lunar eclipses have sparked **myth, fear, and wonder**. In ancient times, many cultures saw eclipses as omens or divine messages. The Incas believed a jaguar was attacking the Moon. Chinese myths spoke of celestial dragons devouring it. Meanwhile, in Hindu mythology, **Rahu and Ketu**, the shadow entities, are said to cause eclipses as part of a cosmic vendetta.

Today, eclipses serve not just as astronomical events, but as opportunities to **connect with cultural heritage and science alike**.

A lunar eclipse is more than just an astronomical alignment — it's a moment to **pause, look up**, and remember that we are part of something vast, beautiful, and in motion.

A total lunar eclipse, also known as a "Blood Moon," will occur overnight on September 7–8, 2025. The eclipse will be visible across Asia, Australia, Africa, and Europe.

-Sanjana Milind
Amateur Scientist, Skygaze India

EXOPLANETS: OTHER WORLDS BEYOND OUR SOLAR SYSTEM

Have you ever looked at the night sky and wondered — are **we alone in the universe?** Could there be other Earth-like planets out there, orbiting stars just like our Sun?

The answer is: **Yes, maybe!**

Scientists have discovered **thousands of planets** that exist **outside our solar system**. These are called **exoplanets** — short for “**extrasolar planets**”. They are planets that orbit **other stars**, not our Sun. And guess what? Some of them might even be **habitable**, like Earth!

An **exoplanet** is any planet that exists **outside our solar system**, meaning it orbits a star that is **not the Sun**. Just like our Earth orbits the Sun, these planets orbit their own stars — which could be bigger, smaller, hotter, or cooler than our Sun.

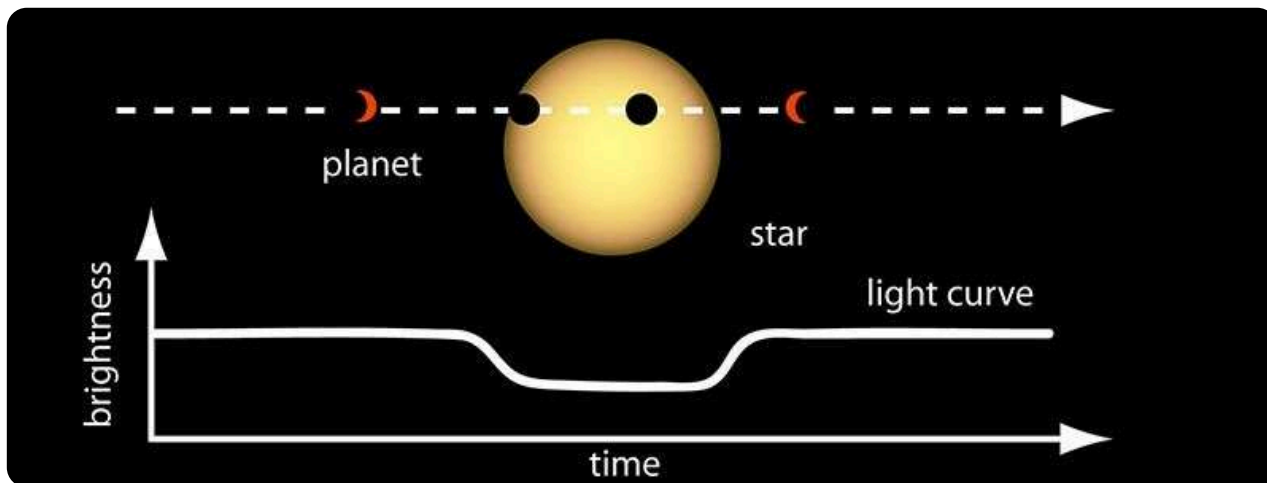
Some exoplanets are **rocky** like Earth or Mars. Others are **gas giants** like Jupiter or Saturn. And some are even **weird and wild** — imagine planets where it rains molten glass sideways or where oceans are made of lava!

We cannot directly see most exoplanets, even with powerful telescopes. Why? Because stars are **too bright**, and planets are **too small and dim** to be spotted easily.

So, astronomers use **clever techniques** to **indirectly detect** exoplanets.



Exoplanets



Transit Method

1. The Transit Method

Imagine a small planet passing in front of its star — like a tiny shadow crossing a giant lightbulb. When this happens, the light from the star **dips slightly**. If the dip repeats regularly, scientists know a planet is orbiting that star.

This is the most common method used today. The **Kepler Space Telescope** used it to discover thousands of planets.

2. The Wobble Method (Radial Velocity)

Planets don't just orbit stars — they also cause the stars to move just a little bit, in a small circle or wobble. This movement is tiny, but with sensitive instruments, we can measure it.

This method helps scientists figure out the **mass** of a planet and how far it is from its star.

3. Direct Imaging

This is rare, but sometimes scientists can **actually take pictures** of planets far away, especially when the star isn't too bright or the planet is very big. They use special tools to block the star's light and focus on the planets.

4. Gravitational Microlensing

This is a unique method. When a star passes in front of another distant star, it bends the light from behind it. If the closer star has a planet, it adds a small "blip" in the light pattern. This tells us that a planet is there — even if we never see it.

-Harsh

Founder, Skygaze India

CONTRIBUTE TO BEYOND THE HORIZON!

Have an astronomy article, story, or thought to share?
We're inviting written submissions for our next publication!

Submit via email on support@skygazeindia.com





Skygaze India Brings the Wonders of the Sun to BioEnergy Global Expo 2025 at Yashobhoomi

At the BioEnergy Global Expo 2025 in Yashobhoomi, New Delhi, Skygaze India offered visitors an unforgettable journey into solar science. For three consecutive days, attendees safely observed the Sun using advanced telescopes and solar glasses.

Delhi, India Aug 19, 2025 (issuewire.com) - The recently concluded **BioEnergy Global Expo 2025**, held from 29 to 31 July at Yashobhoomi, New Delhi, witnessed a unique convergence of science, sustainability, and public engagement through the presence of **Skygaze India**, an initiative in space science education and outreach. Over the course of three days, Skygaze India drew thousands of visitors to its solar observation setup, offering them a rare opportunity to witness the Sun through high-powered telescopes and solar glasses.

The experience went beyond the spectacle of safely viewing the solar surface. Visitors were guided through interactive explanations of solar phenomena, the science behind eclipses, and the broader role of the Sun in sustaining life and





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Skygaze India: Pioneering Space Science Education in India

₹17.29

 by Shivam Madaan — August 12, 2025 in Business Reading Time: 2 mins read AA 0



Skygaze India Showcases Solar Science at BioEnergy Global Expo 2025, Yashobhoomi

Posted August 22, 2025 by SkygazeIndia

The BioEnergy Global Expo 2025, held from July 29 to 31 at Yashobhoomi, New Delhi, featured a remarkable blend of science and sustainability through the participation of Skygaze India.

New Delhi, July 31, 2025 — Visitors at the BioEnergy Global Expo 2025, held from July 29 to 31 at Yashobhoomi, New Delhi, were treated to a rare and engaging encounter with the Sun, courtesy of Skygaze India, a leading initiative in astronomy education and outreach. Over three

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Skygaze India Brings the Cosmos Closer at Kunskapsskolan International School

Submitted by SkygazeIndia on Tue, 08/19/2025 - 22:29
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Students at Kunskapsskolan International School, Gurgaon had a rare opportunity to explore the night sky when Skygaze India organized a stargazing session on campus. The event turned the open skies into a science classroom, with over 40 participants observing celestial objects.

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Skygaze India Soon to Launch Skygaze Academy: Simplifying Science for All

Join Skygaze Academy and transform your curiosity into understanding. The goal is to provide a clear and engaging path for curious minds, from young students to lifelong learners, to build a deep and lasting understanding of the scientific world.

THIS MONTH IN INDIAN SPACE HISTORY

SEPT, 1993

INSAT-2B LAUNCH ANNIVERSARY

23

The INSAT-2B satellite was launched to strengthen India's communication and meteorological capabilities. It became a vital part of the INSAT series, serving millions with weather forecasting, cyclone tracking, and telecommunication services.

SEPT, 2019

CHANDRAYAAN-2 ORBITER CONTINUES ITS LEGACY

22

On 7th September 2019, the Vikram lander of Chandrayaan-2 attempted its historic soft landing on the Moon's south pole. Although the lander lost communication during descent, the orbiter remained fully functional and continues to send valuable data to this day.

This moment, though bittersweet, showcased ISRO's bold vision, and the lessons from it directly led to the success of Chandrayaan-3 in 2023.

SEPT, 2023

ADITYA-L1 LAUNCH SUCCESS

02

ISRO made headlines with the successful launch of Aditya-L1, India's first solar mission. The spacecraft is on its way to Lagrange Point 1, where it will study the Sun's outer atmosphere (corona), solar winds, and space weather.

LAUNCHING
SOON



SKYGAZE ACADEMY

At Skygaze Academy, we're making science simple. As a next-generation educational platform, powered by Skygaze India, we harness the power of AI and IT to demystify complex subjects, with a special focus on space science. Join Skygaze Academy and transform your curiosity into understanding.

SKYGAZE
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